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Tectonic deformation of the southwestern part of the Yangtze craton inferred from paleomagnetism

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Abstract

Upper Cretaceous red sandstones have been sampled at 21 sites from the Jiangdihe Formation in Yuanmou (25.9°N, 101.7°E), the northern part of Yunnan, China. Characteristic directions with unblocking temperature of 680°C have been isolated from all the sites, and provide a positive fold test which is significant at the 99% probability level. Remanent magnetization of the prefolding origin represents a characteristic direction of the Jiangdihe Formation ($D = 26.9°$, $I = 35.6°$, and $\alpha_{95} = 3.6°$). This paleomagnetic direction confirms the fairly large easterly declination of the Cretaceous time for the southern part of the Yangtze craton. Incorporating this result into previous data from the areas along the longitude of 102°E in the southwestern part of Yangtze craton, the declination value in the Cretaceous paleomagnetic direction monotonously decreases from 45 to 2° as latitude increases from 25.0 to 30.1°N. The tectonic rotation expected from the apparent polar wander path of Eurasia shows that the clockwise rotation decreases gradually from 30 to −10°, whilst increasing in latitude. The systematically differential rotation along the longitude in the southwestern part of the Yangtze craton is a part of the deformation mode of the Asian continent due to the collision with India. The Yangtze craton behaved as a unit continuum material with a ductile nature during deformation. © 1998 Elsevier Science B.V. All rights reserved.

Keywords: plate collision; deformation; Asia; Yangtze Platform; paleomagnetism

1. Introduction

The collision of India and Asia is one of the most prominent tectonic events in Cenozoic time. The Indian continent collided with Asia at about 52 Ma and continued to penetrate into the Asian continent [1]. The northward penetration of India gave rise to tectonic deformation in the Asian continent. The deformation feature within the Asian continent has been studied by both laboratory and numerical experiments [2–8]. Latitudinal shortening in front of India and tectonic rotation along both its sides are expected in the Asian continent.

Paleomagnetic evidence on clockwise tectonic rotation has been discovered from the Cretaceous rocks in southwest Asia which flank the northeast corner of India. Easterly declination ranging between 42 and 116° is reported from the Qiangtang, Shan-Thai and Simao terranes [9–13] (Fig. 1). The clockwise rota-

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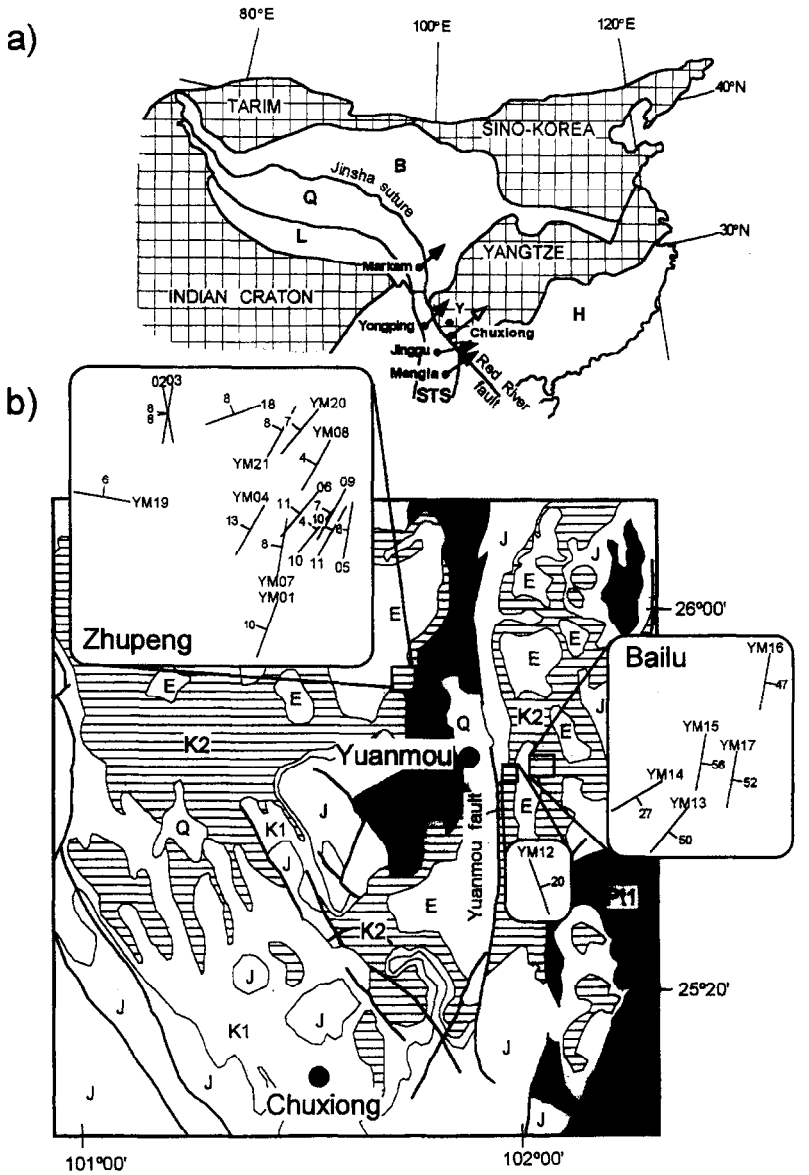


Fig. 1. (a) Sketch tectonic map of Southeast Asia (modified from Enkin et al. [30], and Yang and Besse [52]) and location of study area (Y = Yuanmou). B = Bayan Kala terrane; Q = Qiangtang terrane; L = Lhasa terrane; STS = Shan-Thai and Simao terranes; H = Huanan Fold Belt. Also indicated are main locations in Qiangtang, Shan-Thai and Simao terranes where Cretaceous paleomagnetic data are available. The declination is indicated by arrows [9–13]. Open arrow is for Chuxiong in the Yangtze block [14]. (b) Geological map of study area [20,21]. $Pt1$ = Proterozoic; J = Jurassic; $K1$ = Lower Cretaceous; $K2$ = Upper Cretaceous; E = Paleogene; Q = Quaternary. Strike and dip of strata for each sampling sites is shown in insets.

tion associated with the collision with India occurred indisputably in the eastern part of the Asian continent. This deformation feature is consistent with models predicted by the laboratory and numerical experiments.

Another outstanding easterly paleomagnetic direction is observed in the southern margin of the Yangtze block (Fig. 1). Paleomagnetic study claimed that the Chuxiong area, 60 km northeast of the southwestern boundary of the Yangtze block, was

subjected to clockwise rotation through 45° since the Cretaceous time, with respect to the central part of the Yangtze block [14]. This implies that the Yangtze block was also subjected to tectonic deformation due to collision with India.

The present investigation aims to detect tectonic deformation on the southwestern part of the Yangtze block from the paleomagnetic viewpoint. We chose the Yuanmou area as our paleomagnetic study and focused on the Upper Cretaceous redbeds of the Jiangdihe Formation, because late Mesozoic formation preserves evidence of deformation later than 52 Ma when India collided.

2. Geological setting and sampling

The Yangtze block is located to the east of the eastern margin of Tibet. Although many large earthquakes have recently occurred on the southwestern part of the Yangtze block [15–17], this block is recognized as a craton. Basement of this block is composed mostly of the Proterozoic, which is widely overlain by the Sinian system (850–600 Ma) [18,19].

The Yangtze craton is bounded on the southwest by the Shan-Thai and Simao terranes; the NW to SE trending Red River fault marks the boundary of these two blocks. The terrestrial Jurassic to Cretaceous redbeds widely cover the southwestern part of the Yangtze craton from the Sichuan basin to the Chuxiong basin and extend southwards to the Simao depression of the northern part of the Simao terrane [20,21]. The redbeds are folded by NW to NNW trending anticlines [22]: the folding took place in the Early Tertiary as evidenced by an unconformity between the Upper Eocene–Oligocene rocks and Cretaceous–Paleocene rocks [21].

The Yuanmou area (25.9°N, 101.7°E) is located about 120 km northeast of the Red River fault in the Chuxiong basin on the Yangtze craton (Fig. 1). The total thickness of the Jurassic through Eocene redbeds sequence is over 6 km in this basin [20–22]. This sequence is cut by several north–south striking faults which are southern splays of the Xianshuihe–Xiaojiang system. The Jiangdihe Formation (Jiangdihe 3 and Jiangdihe 2 Formations in ascending order) is the Upper Cretaceous series, made up of purple-red, grayish green sandy mudstone and shale, containing elements of *Cypridea* (*Ulwellia*) houmafangensis and *Manchurichthys* sp. We collected samples of red sandstones in the Jiangdihe Formation at 21 sites from two localities of both sides of the Yuanmou fault: fifteen sites are from Zhupeng (25°55′N, 101°45′E), where strata dip gently westward and the remaining six sites are from Bailu (25°39′N, 102°03′E), where strata dip fairly steeply eastward. Eight to ten hand samples, oriented with a magnetic compass, were distributed over a distance of up to 15 m in a site. Localities of sites were determined by a portable navigational instrument (SONY PYXIS). The present declination value at the sampling point was estimated from the calculation using the International Geomagnetic Reference Field of 1994 [23].

3. Paleomagnetism

In the laboratory, individual specimens, 25 mm in diameter and 23 mm long, were prepared from samples. Natural remanent magnetizations (NRMs) were measured with a 2-G cryogenic magnetometer. One specimen from each sample was subjected to stepwise thermal demagnetization up to 690–710°C. Results for each specimen were plotted on orthogonal vector diagrams [24] to assess component structure as well as on equal-area projections to evaluate directional stability. Principal component analysis [25] was used to estimate the component directions (Table 1).

Acquisition of saturation isothermal remanent magnetization (IRM) and back-field demagnetization were performed on a 2-G pulse magnetizer up to maximum fields of 2.7 T (Fig. 2a,b). High saturation field more than 2.7 T and high coercivity of the remanence of 400–600 mT indicate that hematite is major magnetic carrier for IRM of red sandstones in the Jiangdihe Formation. This is confirmed by thermal demagnetization of the composite IRM [26] (Fig. 2c,d).

3.1. Demagnetization

Demagnetization behavior strongly depends on NRM intensity of a sample. Samples with weaker

Table 1
Paleomagnetic results for redbeds in the Cretaceous Jiangdihe Formation from Yuanmou, Yunnan in the Yangtze craton

Site	n/N	NRM (10^{-3} A m^2)	In situ		Tilt-corrected		k	α_{95} (°)	Locality		Formation
			Dec. (°)	Inc. (°)	Dec. (°)	Inc. (°)			Latitude (N)	Longitude (E)	
High-temperature component											
Zhupeng											
YM01	10/10	5.4	32.7	34.5	25.6	36.0	177.7	3.6	25°54.7′	101°45.4′	Jiangdihe 3
YM02	9/9	6.5	26.5	37.3	20.8	41.4	123.9	4.6	25°56.0′	101°44.8′	Jiangdihe 3
YM03	8/8	5.7	32.5	25.3	28.8	28.6	42.4	8.6	25°56.0′	101°44.8′	Jiangdihe 3
YM04	9/9	9.2	36.2	36.8	26.4	37.1	94.3	5.3	25°55.3′	101°45.3′	Jiangdihe 3
YM05	8/8	4.6	32.1	23.1	28.6	25.5	55.0	7.5	25°55.3′	101°45.9′	Jiangdihe 3
YM06	6/9	4.5	28.7	24.1	24.2	21.3	149.6	5.5	25°55.4′	101°45.6′	Jiangdihe 3
YM07	10/10	6.1	39.5	37.4	33.6	40.7	46.2	7.2	25°55.2′	101°45.5′	Jiangdihe 3
YM08	10/10	7.8	31.3	38.1	28.1	38.0	67.0	5.9	25°55.7′	101°45.7′	Jiangdihe 2
YM09	9/9	7.1	31.7	37.5	26.3	37.5	266.5	3.2	25°55.4′	101°45.8′	Jiangdihe 2
YM10	9/9	2.1	30.4	40.3	23.0	39.2	60.2	6.7	25°55.3′	101°45.7′	Jiangdihe 2
YM11	9/9	6.3	24.9	40.2	16.8	38.5	23.9	10.7	25°55.3′	101°45.8′	Jiangdihe 2
YM18	10/10	5.8	27.5	37.1	23.4	31.7	112.3	4.6	25°56.0′	101°45.2′	Jiangdihe 3
YM19	10/10	9.6	28.1	33.9	26.9	28.3	43.1	7.4	25°55.5′	101°44.4′	Jiangdihe 3
YM20	10/10	7.9	27.1	29.0	23.3	27.9	108.8	4.7	25°55.9′	101°45.6′	Jiangdihe 3
YM21	10/10	8.6	32.3	41.1	25.3	41.6	70.6	5.8	25°55.9′	101°45.5′	Jiangdihe 3
Mean direction	15/15		30.8	34.4			141.8	3.2			
					25.4	34.3	127.7	3.4			
Bailu											
YM12	10/10	4.2	5.6	58.6	26.7	46.2	85.4	5.3	25°38.2′	101°58.3′	Jiangdihe 2
YM13	9/9	8.1	353.1	27.6	33.7	51.0	166.4	4.0	25°38.9′	102°02.6′	Jiangdihe 3
YM14	9/9	4.7	34.0	29.0	51.0	36.4	125.9	4.6	25°39.1′	102°02.4′	Jiangdihe 3
YM15	10/10	7.6	339.7	24.7	16.6	37.9	37.0	8.1	25°39.3′	102°02.8′	Jiangdihe 3
YM16	10/10	7.9	355.9	31.3	24.7	28.4	87.7	5.2	25°39.8′	102°03.2′	Jiangdihe 3
YM17	10/10	6.0	355.4	35.2	31.9	30.4	116.7	4.5	25°39.2′	102°03.0′	Jiangdihe 3
Mean direction	6/6		359.9	35.5			16.9	16.8			
					30.7	38.9	40.8	10.6			
Mean direction of all sites									25°54.0′	102°42.0′	
	21/21		22.4	35.5			25.3	6.4			
					26.9	35.6	77.9	3.6			
Middle-temperature component											
YM12	8/10		264.0	−10.0			17.4	13.7	25°38.2′	101°58.3′	
					263.7	9.7	16.8	13.9			
Others	27/188		206.2	−32.9			5.3	13.4	25°54.0′	102°42.0′	
					215.8	−28.8	5.0	13.8			
Low-temperature component											
	109/198		347.1	50.8			7.6	5.3	25°54.0′	102°42.0′	
					358.3	50.9	5.5	6.4			

The magnetization directions of high-, middle- and low-temperature components are listed. N and n are numbers of field-oriented samples and samples used for statistics, respectively. NRM = mean intensity value of remanent magnetization of samples for each site before demagnetization; Dec. = declination; Inc. = inclination; k = Fisherian precision parameter; α_{95} = the radius of cone of 95% confidence.

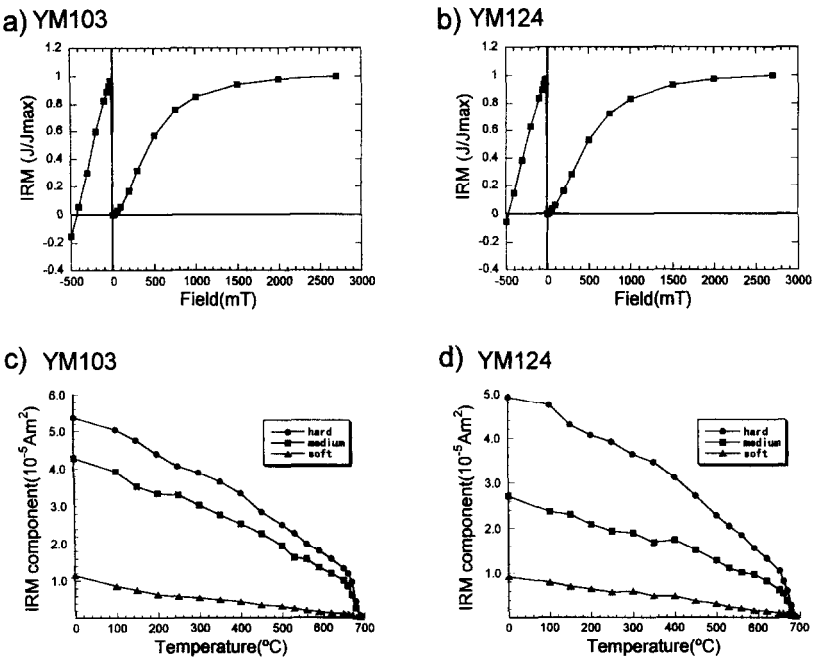


Fig. 2. IRM acquisition curve, back-field demagnetization and thermal demagnetization of IRM for samples (YM103 and YM124) from sites YM10 and YM12. (a,b) Samples acquire over 80% of their IRM in 1 T but do not reach saturation IRM in 2.7 T in DC field. Coercivity of the remanence ranges between 400 and 600 mT. (c,d) Thermal demagnetization of a three-component IRM produced by applying a different DC field (2.7, 0.4 and 0.12 T) to each of three perpendicular axes of a sample.

NRM intensity reveal multi-component magnetization.

Samples with NRM intensity of 2.1–0.86 × 10^{-7} A m^2 (mean intensity, 0.93 ± 0.54 × 10^{-7} A m^2) (sites of YM04, YM10, YM21) revealed a single, well-defined component of magnetization during progressive thermal demagnetization (Fig. 3a,b). Little change is observed in the direction of remanent magnetization up to the Curie temperature of hematite.

Samples with NRM intensity of 0.96–0.54 × 10^{-7} A m^2 (0.63 ± 0.25 × 10^{-7} A m^2) from 10 sites generally had two-component magnetization (Fig. 3c,d). The low-temperature component is removed in the initial stage of demagnetization by 300°C. Unblocking temperature of the high-temperature component ranges between 690 and 710°C, indicating that hematite carries the stable remanent magnetization.

Definite three-component magnetization is observed in 31 samples of eleven sites (YM02, YM03, YM05, YM06, YM09, YM12, YM13, YM14, YM15, YM16, YM17) with NRM intensity 0.81–0.42 × 10^{-7} A m^2 (0.53 ± 0.20 × 10^{-7} A m^2) (Fig. 3e–g). After the low-temperature component is erased by 300°C, the middle-temperature component appears at the temperature range of 150–400°C. Remanent magnetization of this component reveals a reversed direction. The high-temperature component appears in the temperature range of 450–710°C, indicating that the high-temperature component resides in hematite.

Visual inspection provides an indication of the presence of the middle-temperature component magnetization in many samples during the temperature range of 200–450°C, although these samples are categorized as one- or two-component magnetization by the principal component analysis (e.g. Fig. 3c,d). The high-temperature component with a normal polarity is superimposed on the reversed direction of the middle-temperature component. Appearance of the sign of the latter component seems to be controlled by the strength of the high-temperature component magnetization. This middle-temperature component magnetization is probably common for the Jiangdihe Formation. The high-temperature component is, however, extracted, after the obscure sign

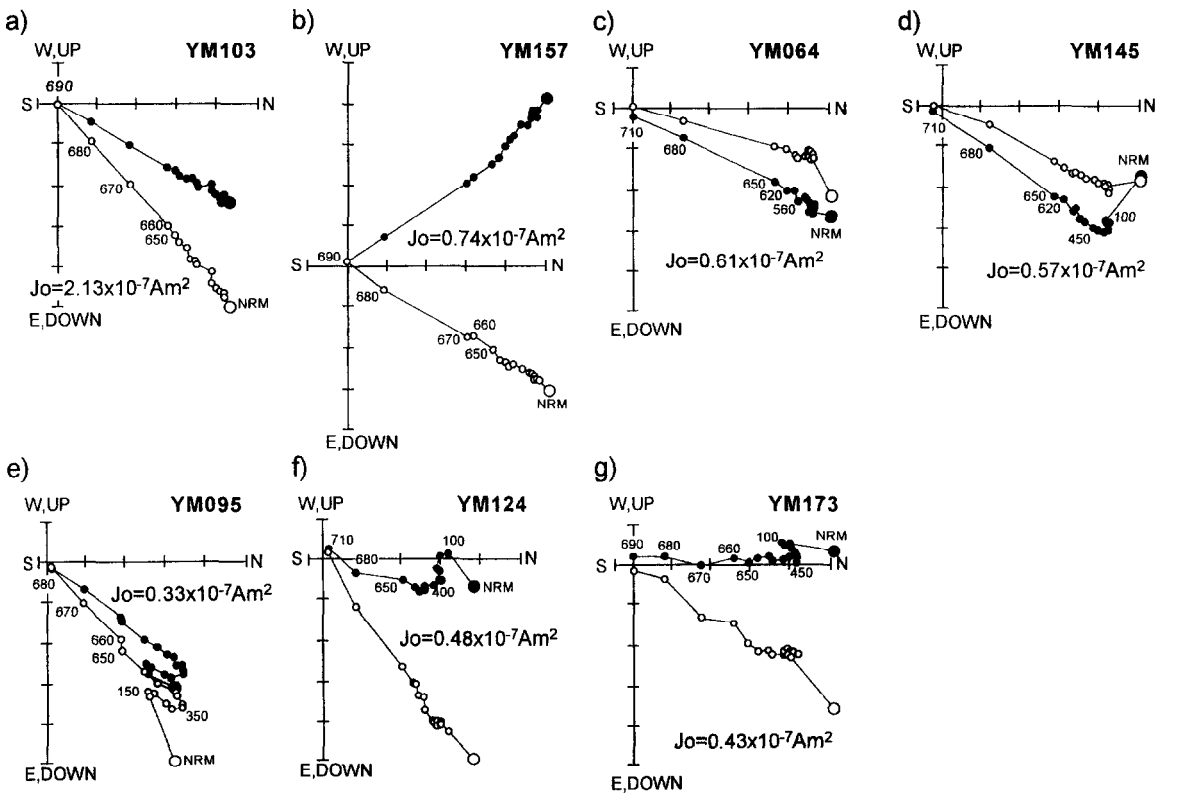


Fig. 3. Orthogonal projections of magnetization vector end points during thermal demagnetization experiments for three categories: single-component magnetization (a,b; samples YM103 and YM157 from sites YM10 and YM15, respectively); two-component magnetization (c,d; samples YM064 and YM145 from sites YM06 and YM14, respectively); and three-component magnetization (e,f,g; samples YM095, YM124 and YM173 from sites YM09, YM12 and YM17, respectively). Solid symbols represent horizontal projection, open symbols represent vertical projection. Tectonic tilt correction has not been applied.

of the middle-temperature component magnetizations vanishes by demagnetization around 450°C.

3.2. Paleomagnetic direction

3.2.1. High-temperature component

The highest temperature component directions are clustered tightly in each site (Fig. 4a): the precision parameter ranges between 23.9 and 266.5.

The site mean paleomagnetic directions for 15 sites from Zhupeng reveal a tight cluster. A tilt corrected mean direction is $D = 25.4°$, $I = 34.3°$, $\alpha_{95} = 3.4°$. No fold test could be performed, because bedding attitudes are almost uniform. Paleomagnetic directions from two formations (Jiangdihe 2 and Jiangdihe 3) are statistically identical.

The site mean paleomagnetic directions for 6 sites from Bailu also reveal a tight cluster. After tilt correction, the site mean paleomagnetic directions are clustered more tightly ($D = 30.7°$, $I = 38.9°$, $\alpha_{95} = 10.6°$, $k = 40.8$) than those in situ ($D = 359.9°$, $I = 35.5°$, $\alpha_{95} = 16.8°$, $k = 16.9$).

Paleomagnetic directions from Zhupeng and Bailu are apart by 25.2° in situ, whereas they approach after tilt correction (angular difference = 6.3°). The precision parameter k increases from 25.3 to 77.9. The fold test is positive at the 99% confidence limit [27]. The Upper Cretaceous Jiangdihe 2 and 3 formations preserve paleomagnetic direction of prefold origin ($D = 26.9°$, $I = 35.6°$, $\alpha_{95} = 3.6°$).

3.2.2. Low-temperature component

Paleomagnetic directions of the low-temperature component magnetization removed by 300°C provide a negative fold test at the 95% confidence limit [27], and are dominated in situ by a component in

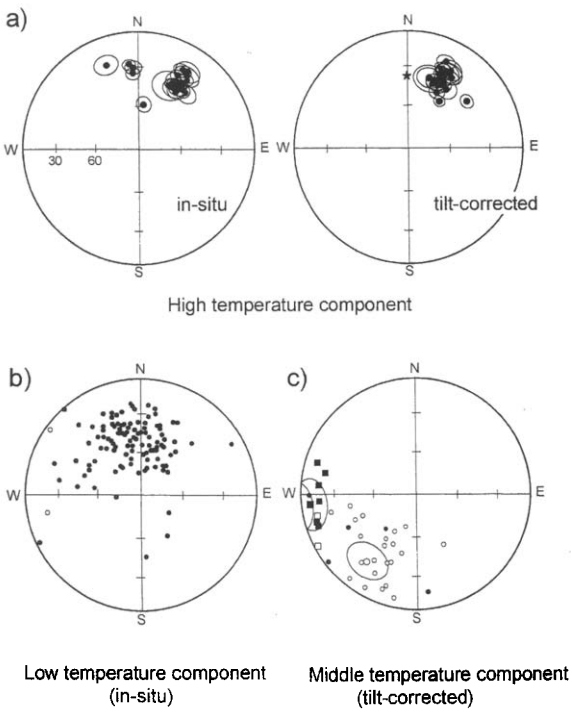


Fig. 4. Equal-area projections of paleomagnetic directions of the Jiangdihe Formation. (a) Equal-area projections of site mean directions of the high-temperature component with a circle of 95% confidence before and after tilt correction. (b) Low-temperature component before tilt correction. (c) Middle-temperature component after tilt correction. Paleomagnetic directions for site YM12 are shown by squares and paleomagnetic directions for the remaining samples are shown by circles. A site mean direction for site YM12 and a mean direction for the remaining samples are shown with a circle of 95% confidence. The solid symbols refer to the lower hemisphere, the open symbols refer to the upper hemisphere.

the present direction of the Earth's magnetic field (Fig. 4b). This component is probably ascribable to viscous partial thermoremanent magnetization acquired in the recent times [28].

Two groups are identified in paleomagnetic directions of the middle-temperature component magnetization (Fig. 4c): eight samples of YM12 reveal a westerly direction (D = 264.0°, I = −10.0°, α_{95} = 13.7° in situ), whereas the remaining 27 samples from other ten sites show a southerly direction before and after tilt correction (D = 206.2°, I = −32.9°, α_{95} = 13.4° vs. D = 218.8°, I = −28.8°, α_{95} = 13.8°). Two distinct directions indicate that the middle-temperature component was probably acquired during two different events. The southerly direction is nearly antipodal to the high-temperature component one. This component presumably originated before the Yuanmou area was subjected to tectonic rotation.

4. Discussion

The remanent magnetization of the high-temperature component after tilt correction for the Jiangdihe Formation represents a characteristic paleomagnetic direction for the Yuanmou area: an easterly deflected declination is observed (D = 26.9°, I = 35.6°, and α_{95} = 3.6°). Reliability of primary origin for the remanent magnetization is ascertained through thermal demagnetization and positive fold test. The prevalence of normal polarity suggests that the characteristic remanent magnetization was probably acquired during the Cretaceous Long Normal Superchron [29].

4.1. Comparison with other paleomagnetic data from the southern part of the Yangtze craton

Cretaceous paleomagnetic directions have been reported from four areas (Ya'an, Xichang, Huili, and Chuxiong) along the longitude of 102°E in the southwestern part of the Yangtze craton (Table 2 and Fig. 5): paleomagnetic results are all from studies of continental Cretaceous redbeds, and almost all the high-temperature component magnetization passed the fold test [9,14,30–32].

The clockwise deflected declination of the Yuanmou area is comparable with the easterly Cretaceous paleomagnetic directions of Chuxiong (25.0°N, 101.5°E) and Huili (26.5°N, 102.4°E) which are the nearest localities to the study area. The Chuxiong area shows an easterly declination of about 44.6° in the Upper Cretaceous Matoushan and Puchanghe Formations [14]. Huang and Opdyke [32] reported a declination of 8.1° for the Upper Cretaceous Xiaoba Formation and 22.1° for the Lower Cretaceous Feitianshan Formation from Huili area. Because the latter did not pass the fold test, a Cretaceous characteristic direction is evaluated by recalculation from 25 sites of the Upper and Lower Cretaceous Formations; D = 12.1° (α_{95} = 5.1°). The Cretaceous

Table 2
Paleomagnetic rotation calculated for localities in the southwestern part of the Yangtze craton and surrounding areas

Locality	Latitude (°N)	Longitude (°E)	Rock-age	Observed direction			VGP			Rotation $R \pm \Delta R$ (°)	Latitude anomaly $\lambda ob - \lambda ex \pm \delta\lambda$ (°)	Reference
				Declination (°)	Inclination (°)	α_{95} (°)	Latitude (°N)	Longitude (°E)	α_{95} (°)			
Yangtze craton												
Ya'an (1)	30.1	103.0	K1, K2	2.1	23.4	11.3	76.3	274.5	11.1	-13.0 ± 5.1	-12.0 ± 11.6	[9]
Ya'an (2) (Feixian Guan)	30.0	102.9	Ju–E	2.0	34.2	3.6	78.6	273.4	3.7	-13.1 ± 9.8	-5.3 ± 5.0	[30]
Xichang	27.9	102.3	K1	3.4	41.2	5.3	84.8	245.8	5.7	-12.2 ± 10.2	1.6 ± 6.6	[31]
Hulli	26.5	102.4	K1, K2	12.1	38.4	5.1	78.0	212.8	5.5	-3.3 ± 9.9[a]	0.8 ± 6.4	[32]
Yuanmou	25.9	101.7	K2	26.9	35.6	3.6	64.6	199.6	3.3	12.4 ± 10.0	0.0	This study
Chuxiong	25.0	101.5	K1	44.6	41.3	10.7	49.2	182.0	11.4	29.4 ± 16.2[a]	4.6 ± 11.9	[14]
Shan-Thai–Simao terrane												
Xiaguan	25.6	100.2	K2	6.9	47.7	8.6	83.6	152.7	10.0	-7.5 ± 15.7		[12]
Yongping	25.5	99.5	K1	42.0	51.1	15.7	50.9	167.3	20.6	27.0 ± 26.6[a]		[11]
Jinggu	23.4	100.4	K2	84.4	39.6	17.8	13.6	171.5	–	70.2 ± 25.1		[13]
	23.5	100.8	K1	115.8	36.0	12.1	13.9	341.3	–	100.8 ± 16.8[a]		
Jinggu	23.4	100.9	K1	79.4	43.3	9.1	18.9	170.0	8.9	64.4 ± 14.7[a]		[12]
Mengla	21.6	101.4	K1	60.8	37.8	7.6	33.7	179.3	8.2	45.9 ± 12.3[a]		[12]
Qiangtang terrane												
Markam	29.7	98.9	K	48.2	49.0	8.8	29.7	98.6	9.5	33.4 ± 16.1		[9]
	29.7	98.7	K2	39.7	51.6	8.5	56.7	172.7	10.6	42.4 ± 18.7		[10]
	29.7	98.4	K1	57.7	50.2	10.9	40.6	170.5	13.0	24.9 ± 16.4[a]		

Rock age: J = Jurassic; K1 = Lower Cretaceous; K2 = Upper Cretaceous; E = Paleogene; k = Fisherian precision parameter; α_{95} = the radius of cone of 95% confidence. Rotation is calculated with respect to the declination expected from 100 Ma pole of the Eurasia APWP [34]. The amount of rotation (R) and its uncertainty (ΔR) are given $D_{ob} - D_{ex}$ and $[(\Delta D_{ob})^2 + (\Delta D_{ex})^2]^{1/2}$ where ΔD_{ob} and ΔD_{ex} are the uncertainty at 95% confidence ($\Delta D = \sin^{-1}[\sin(\alpha_{95})/\cos I]$ with α_{95} the radius of the 95% confidence circle and the I the inclination). The latitude anomaly is given $\lambda_{ob} - \lambda_{ex}$, where λ_{ob} is calculated from the paleomagnetic inclination observed at each site and λ_{ex} is expected from the paleomagnetic pole of Yuanmou for each site. The uncertainty of the latitude anomaly is given $[(\Delta\lambda_{ob})^2 + (\Delta\lambda_{ex})^2]^{1/2}$, where $\Delta\lambda_{ob}$ and $\Delta\lambda_{ex}$ are the uncertainty at 95% confidence [51].

[a] Rotation with respect to the declination expected from the pole of 120 Ma.

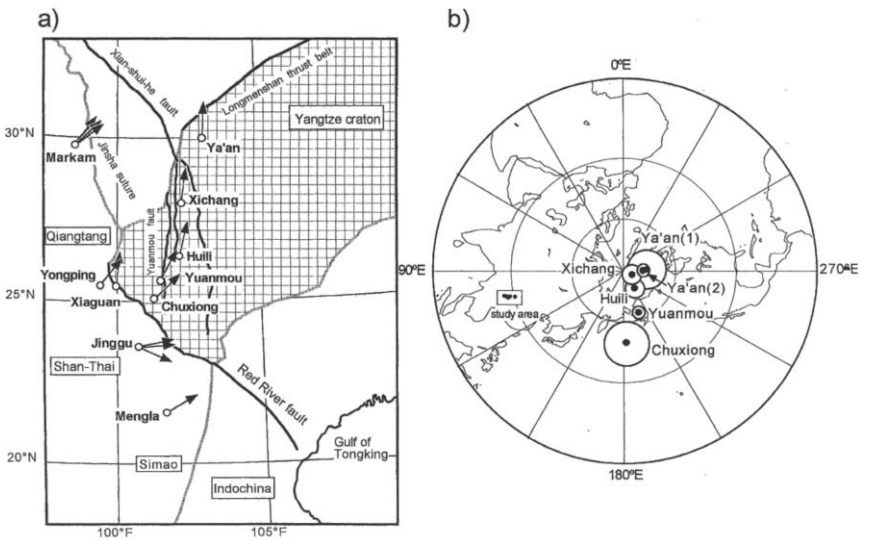


Fig. 5. The declinations of the Cretaceous paleomagnetic directions and Cretaceous paleomagnetic poles (with associated circles of 95% confidence) of the southwestern part of the Yangtze craton. (a) Declinations of the southwestern part of the Yangtze craton and surrounding areas are shown by arrows. (b) Cretaceous paleomagnetic poles of five areas in the southwestern part of the Yangtze craton with associated circles of 95% confidence.

paleomagnetic direction of the Yuanmou falls into intermediate direction between the Cretaceous paleomagnetic directions of Chuxiong and Huili.

The clockwise deflected declination is restricted to the southern part of the Yangtze craton, because little clockwise deflected declination is observed in Xichang and Ya'an which are located to the further north of Huili [9,30,31]. The Cretaceous redbeds from Xichang (27.9°N, 102.3°E) show a declination value of 3.4° [31]. Two data set are reported from the Ya'an area (30°N, 103°E); the Jurassic and Eocene redbeds provide 2.0° in declination, while the Cretaceous redbeds show 2.1° in declination [9,30]. The mean declination for Ya'an is 2.1°. The Ya'an and Huili areas are characterized as northerly paleomagnetic direction.

Detailed observation in declination of the southwestern part of the Yangtze craton suggests a gradual decrease of the declination value toward the north in the Yangtze craton. The declination of Chuxiong, Yuanmou, Huili, Xichang and Ya'an areas is shown as a function of latitude in Fig. 6. The declination value monotonously decreases from 45 to 2° as latitude increases from 25.0 to 30.1°N. The systematic decrease in declination is clearly represented by paleomagnetic pole positions. The paleomagnetic poles are shown in Fig. 5 for the five areas along the longitude of 102°E (Ya'an, Xichang, Huili, Yuanmou and Chuxiong). The two poles from the northern part of the Yangtze craton (Ya'an, Xichang) are around the spin axis, whereas the poles of Yuanmou and Chuxiong are located further south. This shows that the southern part of the Yangtze craton was tectonically rotated clockwise with respect to the northern part of the Yangtze craton. Rotation mode is differential: the southern part is subjected to larger clockwise rotation.

Inclination data imply a small north–south extension in the southwestern part of the Yangtze craton. We calculate latitude anomalies of Chuxiong, Huili, Xichang and YaÕan for the Cretaceous period (Table 2). These are defined as the differences between observed latitude values and expected values from the paleomagnetic pole of Yuanmou. Although the latitude anomalies lead to insignificant and marginally reliable differences, southward translation is suggested for the southern area (Chuxiong; anomaly $= 4.6 \pm 11.9°$) and northward translation is inferred for the northern area (Ya'an; $-5.3 \pm 5.0°$). The north–south extension possibly occurred in the southwestern part of the Yangtze craton since the late Cretaceous. This tectonic event may have been accompanied by paleomagnetic rotation in the south-

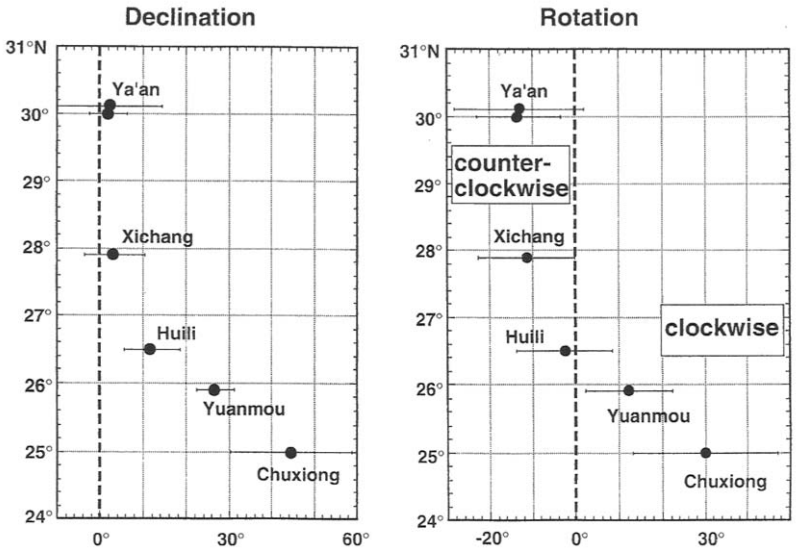


Fig. 6. Plots of declination and amount of rotation with respect to the stable part of the Yangtze craton as a function of latitude for five areas located along the longitude of 102°E in the southwestern part of the Yangtze craton. The error bars show uncertainty at 95% confidence level (Table 2).

western part of the Yangtze craton. An alternative explanation for the low paleolatitude of Ya'an may be the inclination shallowing of laminated sediments in redbeds reported from the Catalan basin of Spain [33], so that further paleomagnetic study is required for the Ya'an area.

4.2. Tectonic deformation mode of the Yangtze craton

The amount of rotation recorded in the southwestern part of the Yangtze craton shows a systematic change in rotation sense from clockwise to counter-clockwise toward the north. The amount of rotation is evaluated with respect to the stable part of the Yangtze craton (Table 2). The expected direction is calculated from the apparent polar wander path for Eurasia [34], because reliable paleomagnetic data in the Yangzte craton are too scarce to construct a characteristic Cretaceous paleomagnetic pole for the stable part of the Yangtze craton [35]. The amount of rotation for Chuxiong, Yuanmou, Huili, Xichang and YaÕan along the longitude of 102°E in the southwestern part of the Yangtze craton is plotted as a function of latitude (Fig. 6). The Chuxiong and Yuanmou areas experienced clockwise rotational motion of more than 12°. The Xichang and Ya'an areas are, on the contrary, subjected to counter-clockwise rotation of more than 12°. The rotation in the clockwise sense gradually decreases from 29 to −13° as latitude increases from 25.0 to 30.1°N. The rotation sense changes at Huili from clockwise to counter-clockwise.

The systematic change in rotation in the southwestern part of the Yangtze craton is not ascribable to local deformation associated with the motion of strike–slip faults (Fig. 7a,b). One mechanisms in this category is motion of a block which is bounded by strike–slip faults [2,16,22,36,37] (Fig. 7b). The dimension of the rotating blocks ranges from an areal size to a continental one. A rotation model of small blocks is postulated by the geological viewpoints [22,38]: since the southern part of the Yunnan area is cut into several rectangular blocks by at least five splays of left-lateral Xianshuihe and Xiaojiang fault system, the clockwise rotation is expected for these blocks associated with left-lateral strike–slip faulting. Clockwise rotation of a whole rigid block of the Yangtze craton is displayed in the laboratory experiments of the plasticine models [2]. These models demand the same amount of rotation within the block. This is not attained by the observation of the paleomagnetic directions which shows differential rotation within the Yangtze craton.

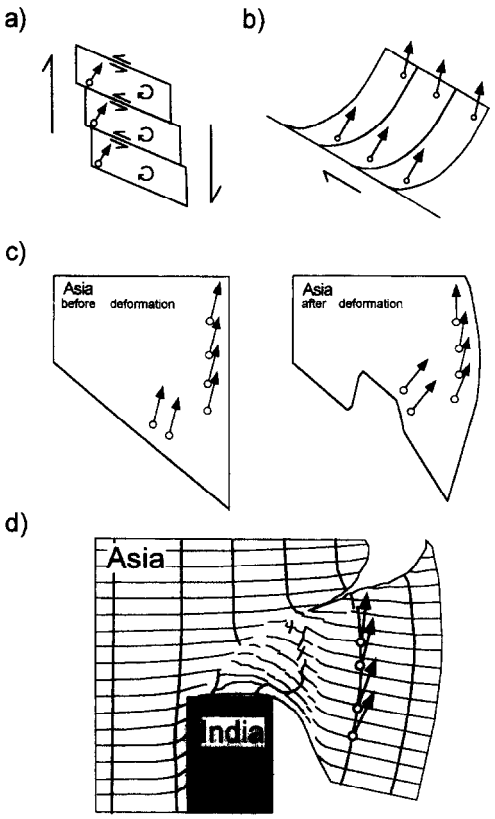


Fig. 7. Possible rotation models in the western part of the Yangtze craton. Paleomagnetic directions (arrows) are superimposed on the models of deformation feature. (a) Clockwise rotation of the blocks between the left-lateral faults [36]. (b) Clockwise rotation associated with plastic drag of the crust due to right lateral movement of a fault [39,40]. (c) Tectonic rotation associated with deformation of the continuum material of the Asian continent due to collision with India [8]. Deformation aspect is calculated for viscous continuum material, assuming that the Asian continent experiences eastward displacement. (d) Tectonic rotation in a model of laboratory deformation experiments for relatively homogenous and isotropic plasticine due to collision with India, allowing for the eastward displacement of the Asian continent [5]. After a whole block of the Asian continent breaks into several smaller blocks, each block keeps a ductile nature as continuum material.

An alternative mechanism of this category is plastic drag of the crust along the fault: right-lateral motion of a fault will give rise to clockwise rotation on areas along the fault, and vice versa [39–41] (Fig. 7a). The Red River fault marks the southwestern boundary of the Yangtze craton and is located about 60 km from Chuxiong. The left-lateral strike–slip motion lasted from 35 to 17 Ma on this fault, whereas the right-lateral motion occurred in the Quaternary [42]. Clockwise rotation of Chuxiong and Yuanmou is not expected from the Tertiary large left-lateral movement. Since recent paleomagnetic studies suggest that crustal rotation due to fault motion is limited within a few kilometers from the fault [43,44], the Quaternary strike–slip motion possibly gives rise to little crustal rotation for these areas. One of the major faults within the Yangtze craton is the left-lateral Xianshuihe and Xiaojiang fault system, but the paleomagnetic study areas in Xichang, Huili and Yuanmou are more than 10 km from this fault system. Paleomagnetic results show clockwise and null rotations as well as counter-clockwise rotation for these areas (Fig. 5), indicating that these variable tectonic rotations are hardly ascribed to only strike–slip motion of the fault system. Any models of this category cannot explain the differential rotation which characterizes deformation aspect of the Yangtze craton.

The other category for the tectonic rotation mode is continental deformation of the continuum material. When the Yangtze craton behaves as a ductile continuum material during the collision with India (Fig. 7c,d), the systematically differential rotational mode along the longitude in the Yangtze craton arises as well as largely clockwise rotation in the Qiangtang, Shan-Thai and Simao terranes. Houseman and England [8] simulate deformation feature of a viscous continuum material when India collided with Asia, assuming that the Asian continent experiences eastward displacement. This approach reveals fairly large clockwise rotation in an area close to India and differential rotation in the Asian continent apart from it (Fig. 7c). Rotation sense clearly changes from clockwise to counter-clockwise toward the north. The similar differential rotation also arises in a model of laboratory deformation experiments for relatively homogeneous and isotropic plasticine allowing the eastward expulsion [5,6,45]: after a whole Asian block breaks into several small blocks, each break-up block preserves a ductile nature as continuum material (Fig. 7d). Differential rotation appears within the block whenever it behaves as a ductile continuum body.

We conclude that the deformation feature observed in paleomagnetic data is not due to local deformation by strike-slip faults, but is a part of the

large-scale continental deformation during the collision with India. The Yangtze craton has been a unit block and has never divided into several blocks since collision with India. The southwestern part of the Yangtze craton probably behaved as a continuum material with a ductile nature.

Our study implies that the Cretaceous redbeds distributed on the southwestern part of the Yangtze craton reflect ductile deformation mode of the lithosphere beneath the upper crust. This suggests that the upper crust apparently behaves as a ductile material over a long interval during the collision with the Indian continent, although the upper crust is believed to behave as a brittle material during the fast deformation event (e.g. [46]). Ductile deformation of the redbeds was possibly accommodated by the well-developed folds with NW–NNW–N trending axes in the southern part of the Yangtze craton [22]. Formation of the blocks surrounded by left-lateral strike–slip faults probably represents the brittle nature of redbeds during the latest event in the late Cenozoic time.

4.3. Lithospheric nature of the Yangtze craton

Systematic differential rotational motion in the southwestern part of the Yangtze craton contrasts with large and variable clockwise rotation in the Qiangtang, Shan-Thai and Simao terranes (Table 2). Clockwise rotation ranging from 25 to 43° is expected for Markam (30°N, 99°E) in the Qiangtang terrane [9,10]. Yongping in the Shan-Thai terrane was rotated clockwise through 28° [11], whereas Xiaguan along the Red river fault rotated counterclockwise through 7° [12]. Large rotation, ranging between 47 and 101°, is observed in the Simao terrane [12,13]. Instead of the systematic rotation in the Yangtze craton, complicated rotational motions seem to have dominated in these terranes close to the Yangtze craton.

The large and variable clockwise rotation is probably ascribed to the soft and deformable nature of the Qiangtang, Shan-Thai and Simao terranes which flank the stiff Yangtze craton. A heterogeneity model in lithospheric strength predicts the complicated rotation mode in the surrounding soft lithosphere of the strong craton [47,48]. The Qiangtang terrane is a Triassic–Jurassic orogenic collage located on the south of the Asian continent. The Shan-Thai and Simao terranes are a tectonic collage of continental fragment and arcs from Gondwana and Cathaysia, which welded to the Yangtze craton in Triassic time [18,49,50]. The newly created continental marginal part is possibly softer than a craton. Although the Yangtze craton exhibits a ductile nature, it was stronger and more resistant to deformation than the surrounding newly created and mechanically weak crust.

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